

Aryaman Sharma

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EDUCATION

University of Waterloo - **STUDENT ID: (21192935)**

Waterloo, Canada

B.C.S. Computer Science Honours Co-op (GPA: 3.70/4.0)

September 2025 – April 2030 (Expected)

- **Awards:** President's Scholarship of Distinction, 1x Term Distinction
- **Coursework:** Elementary Algorithm Design and Data Abstraction, Designing Functional Programs, Calculus 1 & 2, Algebra with Proofs, Linear Algebra 1, Tools/Techniques for Software Development, Introduction to Microeconomics

PROJECTS

Heart Disease Prediction | Python, scikit-learn, NumPy, Pandas, Git **Link:** github.com/arymn7/heart_disease_prediction

- Developed an end-to-end **machine learning pipeline** to predict **heart disease risk**, achieving **~85% classification accuracy**, using **logistic regression** on **300+ patient records**.
- **Improved model reliability** through **data cleaning and feature engineering**, validated via an **80/20 train–test split**, using **Python, Pandas, NumPy, and scikit-learn**.
- Evaluated **model performance** on a held-out test set using **accuracy and classification metrics**, ensuring robust generalization, with **scikit-learn's** evaluation framework.

RC Vehicle | C++, Arduino, Embedded systems, Electronics

- Designed and built an **RC car** using **Arduino, C++ firmware** using **structured, modular code** and **interrupt-based logic**, achieving a top speed of **15 km/h** with **low-latency real-time motor control**.
- Integrated **embedded systems** components for **wireless operation**, delivering a **~30 m control range** with **stable signal performance** and reliable **system responsiveness**.

Portfolio Website | React, JavaScript, HTML/CSS, GitHub Pages

Link: <https://github.com/arymn7/portfolio-website>

- **Designed and developed** a **fully responsive** personal portfolio using **React** and modern **JavaScript**, showcasing projects, experience, and contact details in a **clean UI**.
- **Improved performance and accessibility** through **reusable components, semantic HTML, optimized assets, and mobile-first CSS layouts**, ensuring **smooth cross-device rendering**.
- **Deployed and maintained** the site via **GitHub Pages** with **Git-based version control**, enabling **seamless updates and continuous improvement**.

BlackJack | Python, Pygame, Git

Link: github.com/arymn7/blackjack-pygame

- Built a **fully interactive Blackjack** game with **real-time rendering** and **input handling**, supporting **10+ game states** and complete win/loss/draw resolution, using **Python** and **Pygame**.
- Implemented **robust game logic** with hit/stand actions, automated dealer behavior (hit until 17), score calculation, and bust detection, scaling gameplay to **200+ cards** with **randomized shuffling**, using **object-oriented design**.
- Designed a **user-friendly graphical interface** with **restart/replay functionality**, ensuring **smooth gameplay** and **maintainability**, while managing **iterative development** and **bug fixes** via **Git** with structured commits.

EXPERIENCE

Fletcher's Meadow Secondary School

Toronto, Canada

Peer Tutor - Computer Science & Mathematics

September 2023 – June 2025

- Delivered **150+ hours** of support in computer science, teaching **object-oriented programming** principles in (**Java/Python**) and guiding peers through coding projects, which **boosted coding proficiency** and **problem-solving**.
- **Tutored 15+ peers** in calculus by simplifying **complex concepts** like derivatives, integrals, and limits into accessible explanations, reinforcing students' strategies while leading to **20–30% improvements** in test performance.

SKILLS

Languages: C, C++, Python, Java, HTML, CSS, JavaScript, Racket, SQL

Technologies: Git, Pandas, Scikit Learn, NumPy, Pygame, React, NodeJS

Developer Tools: GitHub, VS Code, IntelliJ IDEA, PyCharm, Linux, Vim, Power BI, Excel

ACHIEVEMENTS

Certificate Of Distinction in the University Of Waterloo's Sir Issac Newton Exam (Physics)

May 2025

Certificate Of Distinction in the University Of Waterloo's Avogadro Exam (Chemistry)

May 2024